

**BEFORE YOU READ — ANSWER FROM MEMORY**

1. What are the two forms of *Entamoeba histolytica*, and which one is infective vs disease-causing?
2. What single microscopic finding inside a trophozoite is considered pathognomonic for *E. histolytica*?
3. Which extraintestinal site is most commonly affected by invasive amebiasis, and via what route does the organism get there?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Etiology and classification

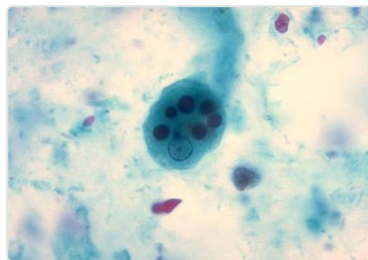
- **Amebiasis** = parasitic infection by the protozoan ***Entamoeba histolytica***, found worldwide
- **Global impact**: affects millions annually, especially in **tropical regions with poor sanitation**; an estimated **40,000-100,000 deaths/year**
- **Transmission**: **fecal-oral route** — contaminated food, water, or surfaces; also **sexual transmission** (oral-anal practices/anilingus); **poor nutrition** (via impaired immunity) is a risk factor

Form	Role
Cyst	Infectious, dormant form — survives outside the host, quadrinucleated, contains glycogen vacuole and chromatoid bodies
Trophozoite	Active, disease-causing form — motile, invasive; survives only hours outside tissue

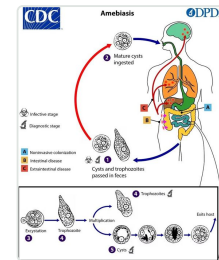
The genus *Entamoeba* has several species in the human gut (*E. histolytica*, *E. dispar*, *E. moshkovskii*, *E. bangladeshi*, *E. polecki*, *E. coli*, *E. hartmanni*). Only *E. histolytica* is definitively pathogenic. *E. dispar* looks identical on microscopy but is a commensal (molecular techniques distinguish them); *E. moshkovskii* is linked to infant diarrhea.



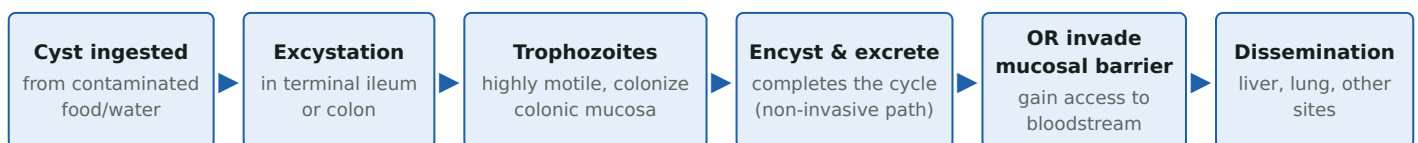
E. histolytica cyst.



E. histolytica trophozoite.

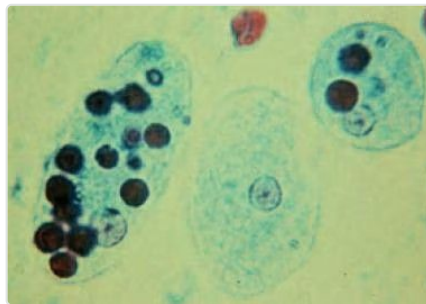


Life cycle: cysts ingested → excystation in terminal ileum/colon → trophozoites colonize or invade.

2 Pathophysiology

E. histolytica is a pseudopod-forming, non-flagellated protozoan causing proteolysis, tissue lysis, and host-cell apoptosis. Humans (and possibly non-human primates) are the only natural hosts. Disease can be caused by only a small inoculum of cysts.

Microscopic feature	Significance
Cytoplasm	Two zones — hyaline outer margin and granular inner region
Ingested RBCs in trophozoite cytoplasm	Pathognomonic for <i>E. histolytica</i> (though reported rarely with <i>E. dispar</i> too)
Nucleus	Fine, regular peripheral chromatin granules + small central karyosome (endosome)



Trichrome stain of trophozoites — two have ingested erythrocytes; all three show small, centrally located karyosomes.

3 Clinical manifestations

Category	Features
Intestinal amebiasis	Diarrhea (may be bloody) , abdominal pain, cramping, fever
Extraintestinal amebiasis	Liver abscess (most common) , pleuropulmonary involvement, rarely brain abscess

- **Amebic liver abscess**: right upper quadrant pain + fever; more common in **younger men** in endemic areas; does **not** typically cause a left-shift in WBC count or elevated bilirubin (unlike pyogenic liver abscess)
- **Non-dysenteric amebiasis** can mimic IBS, inflammatory bowel disease, or diverticulitis; a right-sided colonic mass may mimic cancer
- **Amebic dysentery** can be confused with shigellosis, salmonellosis, or schistosomiasis/ulcerative colitis, but typically has **less frequent, less watery** stools with **tenacious mucus and blood**, and **fewer WBCs** (due to trophozoite lysis of neutrophils)

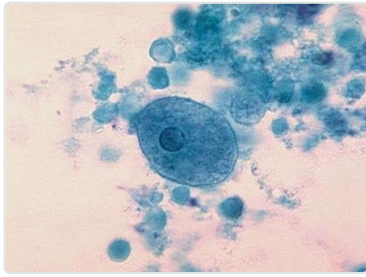
4 Virulence and immune evasion — how it invades

Mechanism	Detail
Adherence	Galactose/N-acetylgalactosamine (GAL/GalNAc)-specific lectin (260-kD surface protein, 170-kD + 35-kD subunits) mediates attachment to colonic epithelium; mucosal IgA against this lectin reduces recurrence
Cytolysis	Amebapores — peptides that form pores in lipid bilayers; also induce apoptosis (sublytic concentrations) via a non-Fas, non-TNF-α pathway
Tissue invasion	Cysteine proteinases — degrade gut tissue, amplify IL-1-mediated inflammation (mimic IL-1-converting enzyme), and cleave/inactivate C3a, C5a, IgA, and IgG (immune evasion)
Cell-mediated immunity	Macrophages kill trophozoites (contact-, oxidative-, nitric oxide-dependent, driven by interferon-γ/6); TNF-α activates neutrophil amebicidal activity; CD8+ T cells show cytotoxic activity in vitro; during acute invasive disease, T-cell response is depressed by a parasite-induced serum factor

5 Diagnosis

- **Pathogenic E. histolytica cannot be morphologically distinguished** from non-pathogenic E. dispar or other species — **this is a key diagnostic limitation**
- Examining **3 stool samples over ≤ 10 days** improves detection

Setting / method	Detail
Intestinal — microscopy	Trichrome-stained fresh stool smear; intracytoplasmic RBCs in trophozoites is diagnostic; stool exam in liver abscess is usually negative
Intestinal — antigen/molecular	Stool ELISA/antigen detection (sensitive and specific); real-time PCR more sensitive than antigen detection; no specific antigen test exists for E. dispar/E. moshkovskii
Intestinal — culture	Fecal/rectal biopsy or liver aspirate; xenic cultivation on modified Locke-egg media ; success rate 50-70% , technically difficult, less sensitive than microscopy
Extraintestinal	Imaging + serology, or a therapeutic trial with an amebicide ; CBC, liver tests, abdominal CT for liver abscess
Antibody detection	ELISA (most common, detects IgG antilectin antibodies); IFA; IHA; IgM also reported; positive in symptomatic intestinal disease and in liver abscess patients symptomatic >1 week — repeat serology 1 week later if initially negative



E. histolytica trophozoites in liver abscess aspirate — trichrome stain.



Gross specimen of intestine damaged by amebic ulcer.

6 Prevention and patient education

- Avoid drinking contaminated water; use bottled water while travelling
- If local water must be used, purify by: **(a)** boiling >1 minute, **(b)** 0.22 µm filtration, **(c)** iodination with tetraglycine hydroperiodide
- Avoid raw fruits/salads that are hard to sterilize; eat only cooked food or self-peeled fruit
- Wash uncooked vegetables and soak in **acetic acid or vinegar for 10-15 minutes**

P Must-remember pearls

- **Cyst** = infective/dormant form; **trophozoite** = invasive/disease-causing form; only a **small inoculum** of cysts is needed to cause disease.
- **Intracytoplasmic RBCs within trophozoites** on microscopy is the pathognomonic finding for E. histolytica.
- E. histolytica and **E. dispar look identical on microscopy** — E. dispar is commensal; only molecular/antigen tests distinguish them.
- Extraintestinal spread → **liver abscess is the most common site**; amebic liver abscess characteristically does **not** cause WBC left-shift or hyperbilirubinemia (unlike pyogenic abscess).
- Virulence: **GAL/GaINAc lectin** (adherence) → **amebapores** (cytolysis) → **cysteine proteinases** (invasion + complement/antibody inactivation).

Q Test yourself

1. A stool microscopy sample shows trophozoites with red blood cells visible inside the cytoplasm. What is the significance of this finding?

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|--|---|
| A. Pathognomonic for Entamoeba histolytica infection | B. Diagnostic of Entamoeba coli, a harmless commensal |
| C. Indicates a mixed bacterial-parasitic infection | D. A nonspecific artifact of poor fixation |

2. A patient from an endemic area presents with right upper quadrant pain and fever. Imaging reveals a liver abscess. Which lab finding would be atypical for amebic (versus pyogenic) liver abscess?

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| A. A left shift in the white blood cell count | B. Elevated liver transaminases |
| C. Right upper quadrant tenderness | D. Fever |

3. Why can pathogenic E. histolytica and nonpathogenic E. dispar not be reliably distinguished by routine stool microscopy?

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| A. They are morphologically identical; only molecular or antigen-based tests can differentiate them | B. E. dispar does not form cysts, making it undetectable |
| C. E. histolytica trophozoites are twice the size of E. dispar | D. E. dispar only infects immunocompromised patients |

4. Which virulence factor of E. histolytica mediates initial adherence of trophozoites to colonic epithelial cells?

- | | |
|---|-----------------------|
| A. Galactose/N-acetylgalactosamine (GAL/GaINAc)-specific lectin | B. Amebapores |
| C. Cysteine proteinases | D. Flagellar adhesins |

5. A patient with chronic non-bloody diarrhea is initially misdiagnosed with irritable bowel syndrome. Stool antigen testing later confirms E. histolytica. Which feature of non-dysenteric amebiasis explains this diagnostic confusion?

- | | |
|---|--|
| A. It can closely mimic IBS, inflammatory bowel disease, or diverticulitis clinically | B. It always presents with massive bloody diarrhea, making it impossible to confuse with IBS |
| C. It exclusively affects the liver, never the intestine | D. It is always asymptomatic and never causes diarrhea |

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1 — A** — Intracytoplasmic RBCs within trophozoites is considered pathognomonic for *E. histolytica*.
- 2 — A** — Amebic liver abscess does not typically cause a left shift in WBC count or elevated bilirubin, unlike pyogenic liver abscess.
- 3 — A** — The two species are morphologically indistinguishable; molecular techniques are required to differentiate them, despite having very different pathogenic potential.
- 4 — A** — The GAL/GalNAc lectin, a 260-kD surface protein, mediates adherence of trophozoites to colonic epithelial cells.
- 5 — A** — Non-dysenteric amebiasis can closely resemble IBS, IBD, or diverticulitis, and a right-sided colonic mass may even mimic cancer, making it a diagnostic challenge.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Microbiology GIT Block · MEDucation by Mattin Majid · GIT SGL 10 slides, 2026-2027
Study alongside the lecture, not instead of it.